COMPACT ANCIENT NAVIER–STOKES DYNAMICS AND INVARIANT MEASURES IN SELF-SIMILAR VARIABLES

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Abstract. We study compact time-translation dynamics generated by bounded ancient solutions of the three-dimensional Navier–Stokes equations in backward self-similar variables. The compactness used here is formulated in the C^∞_{loc} topology: bounded ancient solutions generate compact trajectory hulls, and uniform L^3-tightness is stable under passage to such hulls. On every compact invariant hull, Krylov–Bogolyubov averages yield invariant Borel probability measures. These measures satisfy the stationary statistical weak form of the self-similar Navier–Stokes equations. Passing from this statistical identity to an exact stationary equation for the barycenter is equivalent to the vanishing of an explicit quadratic covariance. Thus the invariant-measure construction is separated from the additional rigidity needed to extract a single stationary profile.

1. Introduction

We consider the incompressible Navier–Stokes equations in backward self-similar variables,

$$\partial_\tau V + (V\cdot\nabla)V + \nabla P = \Delta V - \frac{1}{2}(V + y\cdot\nabla V), \qquad \operatorname{div} V = 0, \tag{1.1}$$

on $\mathbb{R}^3\times\mathbb{R}$. The equation is autonomous in the logarithmic time variable. If V is a solution and $s\in\mathbb{R}$, then

$$V^s(y,\tau) = V(y,\tau+s)$$

is again a solution. We study the compact dynamical system obtained by taking the closure of these time translates in the C^∞_{loc} topology.

The preceding minimal-element reduction identifies compact ancient dynamics as the next object in the reduction: a nonzero normalized class, once closed under local limits and time translation, contains a solution of least $L^\infty(\mathbb{R}^3\times\mathbb{R})$ size. The present paper does not assert that such minimality alone proves recurrence or stationary behavior. Instead it proves the compact dynamical consequences available once one works on a compact trajectory hull.

Two issues must be kept distinct. First, the invariant-measure construction is a compactness argument on the hull: time averages of Dirac masses along an orbit have weak limits, and every such limit is invariant under time translation. Second, the nonlinear term is retained exactly at the statistical level: the stationary identity contains the average of $W\otimes W$, not an unspecified defect measure. When this identity is projected to the barycenter, an explicit quadratic covariance appears. Vanishing of that covariance is an additional rigidity statement, and is the precise bridge from invariant statistical dynamics to a single stationary self-similar profile.

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1.1. **Topological conventions.** All compactness statements are made in the Fréchet topology of local smooth convergence, i.e. the C^∞_{loc} topology. On compact trajectory hulls this topology is metrizable, so sequential compactness and compactness are equivalent. The time-translation maps are homeomorphisms on each invariant hull. These elementary points are used below to pass from orbit averages to invariant measures and from invariant measures to statements about their supports.

For probability measures on a compact metric space we use the usual weak topology, namely convergence against all continuous test functions. The compactness and sequential compactness of the space of Borel probability measures in this topology are standard consequences of the Riesz representation theorem and Banach–Alaoglu; see Billingsley [1]. The time-average construction is the classical Kryloff–Bogoliouboff argument [5].

1.2. **Results imported from the earlier papers.** We use the analytic reduction developed in the preceding papers of the series without reproving it. Paper 0 records the pressure-normalized Caffarelli–Kohn–Nirenberg density, its critical scaling, and the local epsilon-regularity mechanism used to detect nonzero blow-up limits [2, definition "Pressure-normalized critical quantities," proposition "Local finiteness and scaling," and theorem "Caffarelli–Kohn–Nirenberg epsilon regularity"]. Paper I constructs centered bounded ancient Type I profiles from Type I blow-up, fixes the pressure gauges, proves nontriviality, and derives the self-similar equation (1.1) [3, lemma "Pressure gauges and local pressure bounds," proposition "Ancient suitable Type I limit," proposition "Nonzero ancient limit," and lemma "Renormalized equation"].

The present paper begins after those reductions. Its PDE input is the bounded mild ancient theory from Paper III: the Ornstein–Uhlenbeck semigroup formulation, the whole-space Helmholtz–Leray pressure convention, local smoothing, smooth representatives, compactness of bounded families, closure of the mild formulation under locally smooth convergence, invariance under time translation, and closedness under uniform L^3-tightness [4, Section 2, definition "Bounded mild ancient solution," lemma "Local smoothing," remark "smooth representatives," proposition "Local compactness," Section 3, and proposition "Closedness under uniform tightness"]. Every use below of local regularity, pressure reconstruction, compactness of bounded families, or L^3-tightness is a direct invocation of those results.

2. Ancient solutions and trajectory hulls

The terminology in this section is that of Paper III [4, Section 2]. After applying the Helmholtz–Leray projection $\mathbb{P}$, (1.1) is interpreted as

$$\partial_\tau V = \Delta V - \frac{1}{2} y \cdot \nabla V - \frac{1}{2} V - \mathbb{P}\,\mathrm{div}(V \otimes V),$$

and the mild formulation is written with the Ornstein–Uhlenbeck semigroup generated by $\Delta - \frac{1}{2} y \cdot \nabla - \frac{1}{2}$. The semigroup formula, the adjoint L^1 bounds, and the differentiated kernel estimates used to pass to locally smooth limits are those proved in [4, Section 2, Ornstein–Uhlenbeck semigroup and kernel estimates]. We write this semigroup as $S(a)$, $a > 0$.

Definition 2.1 (Bounded mild ancient solution)**.** A vector field $V : \mathbb{R}^3 \times \mathbb{R} \to \mathbb{R}^3$ is a bounded mild ancient solution of (1.1) if

$$V \in L^\infty(\mathbb{R}^3 \times \mathbb{R}), \qquad \mathrm{div}\, V = 0$$

in distributions, and if for every $\sigma < \tau$

$$V(\tau) = S(\tau - \sigma) V(\sigma) - \int_\sigma^\tau S(\tau - s) \mathbb{P}\,\mathrm{div}(V \otimes V)(s)\, ds \tag{2.1}$$

as an identity in the weak-* topology of $L^\infty(\mathbb{R}^3)$.

This is the definition used in Paper III [4, definition "Bounded mild ancient solution"]; it is repeated only to fix notation for the dynamical hulls below.

Definition 2.2 (Locally smooth convergence)**.** A sequence W_n converges locally smoothly to W on $\mathbb{R}^3 \times \mathbb{R}$ if, for every $R, T < \infty$ and every integer $m \geq 0$,

$$W_n \to W \quad \text{in } C^m(B_R \times [-T, T]).$$

We use this locally smooth topology throughout. It is the usual Fréchet topology on $C^\infty_{\mathrm{loc}}(\mathbb{R}^3 \times \mathbb{R})$, and in particular it is metrizable.

Explicitly, one may choose a countable exhaustion by compact cylinders

$$Q_N = \overline{B_N} \times [-N, N], \qquad N \geq 1,$$

and use the seminorms

$$p_{N,m}(W) = \sum_{|\alpha|+j \leq m} \sup_{(y,\tau)\in Q_N} |\partial_y^\alpha \partial_\tau^j W(y,\tau)|.$$

The metric

$$d(W, Z) = \sum_{N,m \geq 1} 2^{-N-m} \frac{p_{N,m}(W - Z)}{1 + p_{N,m}(W - Z)}$$

generates the same topology. Therefore, on any compact set K in this topology, every seminorm $p_{N,m}$ is uniformly bounded on K, and the evaluation maps

$$W \mapsto \partial_y^\alpha \partial_\tau^j W(y,\tau)$$

are continuous for fixed (y,τ). If $B_R \times [-T, T]$ is fixed, then the restriction map

$$W \mapsto W|_{B_R \times [-T,T]}$$

is continuous from K into $C^m(B_R \times [-T, T])$. Hence every integral over a fixed compact spatial set whose integrand is a polynomial in finitely many derivatives of W, multiplied by a fixed smooth compactly supported test function, defines a continuous real-valued function on K. These elementary facts are the only topological input used in the sequel.

Lemma 2.3 (Local smoothing and pressure reconstruction)**.** *Let V be a bounded mild solution of (1.1) on $\mathbb{R}^3 \times (\tau_0, \tau_1)$. Then V has the smooth representative supplied by Paper III. For every compact cylinder $Q \Subset \mathbb{R}^3 \times (\tau_0, \tau_1)$ there is a smooth pressure function P_Q, unique up to addition of a function of time, such that (V, P_Q) satisfies (1.1) pointwise on Q.*

Proof. Local smoothing and uniform C^m-bounds for bounded mild solutions are [4, lemma "Local smoothing"]. The passage from the projected equation to the local pressure P_Q, with the whole-space Helmholtz–Leray pressure normalization and uniqueness up to an additive function of time, is the pressure convention and smooth-representative statement in [4, Section 2 and remark "Representatives and pointwise bounds"]. No additional regularity or decay assumption is introduced here. □

Proposition 2.4 (Compactness of bounded families)**.** *Let V_n be bounded mild ancient solutions of (1.1) satisfying*

$$\sup_n \|V_n\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} < \infty.$$

Then a subsequence converges locally smoothly to a bounded mild ancient solution of (1.1).

Proof. This is the compactness and closedness theorem proved in [4, proposition "Local compactness"]: the local smoothing estimates give Arzela–Ascoli compactness on compact cylinders, and the semigroup/adjoint kernel estimates pass the mild formulation to the locally smooth limit. □

Theorem 2.5 (Compact trajectory hull)**.** *Let V be a bounded mild ancient solution of (1.1). Let K be the closure, in the locally smooth topology, of the family*

$$\{V^s : \ s \in \mathbb{R}\}.$$

Then K is compact. Every $W \in K$ is a bounded mild ancient solution of (1.1), with

$$\|W\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} \le \|V\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})}.$$

Moreover the maps

$$\Theta_t W = W^t, \qquad t \in \mathbb{R},$$

define a jointly continuous group of transformations of K.

If, in addition, V satisfies

$$\sup_{\tau\in\mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} \le L \tag{2.2}$$

and there is $\omega : [1,\infty) \to [0,\infty)$, with $\omega(R) \to 0$ as $R \to \infty$, such that

$$\sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V(y,\tau)|^3 \, dy \le \omega(R) \qquad R \ge 1, \tag{2.3}$$

then every $W \in K$ satisfies the corresponding estimates

$$\sup_{\tau\in\mathbb{R}} \|W(\tau)\|_{L^3(\mathbb{R}^3)} \le L$$

and

$$\sup_{\tau\in\mathbb{R}} \int_{|y|>R} |W(y,\tau)|^3 \, dy \le \omega(R), \qquad R \ge 1.$$

Proof. Every translate V^s satisfies the same mild equation and the same L^∞ bound as V, because (1.1) and (2.1) are autonomous in τ. More explicitly, replacing σ, τ in (2.1) by $\sigma+s, \tau+s$ gives the mild identity for V^s, and the essential supremum is unchanged by the time translation. Therefore every sequence V^{s_n} has, by Theorem 2.4, a locally smoothly convergent subsequence whose limit is a bounded mild ancient solution.

We now prove compactness of the closure K directly. Let (W_n) be an arbitrary sequence in K. Since K is the closure of the orbit, for each n there exists $s_n \in \mathbb{R}$ such that

$$d(W_n, V^{s_n}) < \frac{1}{n},$$

where d is the metric defined above for the locally smooth topology. The sequence (V^{s_n}) has a locally smoothly convergent subsequence $(V^{s_{n_k}})$, say

$$V^{s_{n_k}} \to W \qquad \text{in } C^\infty_{\mathrm{loc}}(\mathbb{R}^3 \times \mathbb{R}).$$

Since $d(W_{n_k}, V^{s_{n_k}}) \le n_k^{-1} \to 0$, the triangle inequality gives $W_{n_k} \to W$ in the same topology. The set K is closed by definition, hence $W \in K$. Thus every sequence in K has a convergent subsequence with limit in K. Because the locally smooth topology is metrizable, sequential compactness is equivalent to compactness, and K is compact.

If $W \in K$, then there are $s_n \in \mathbb{R}$ with $V^{s_n} \to W$ locally smoothly. For every point (y,τ),

$$|W(y,\tau)| = \lim_{n\to\infty} |V(y,\tau+s_n)| \le \|V\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})}.$$

Here the L^∞ norm is the essential supremum, but the smooth representatives supplied by Theorem 2.3 obey the same bound pointwise: if a continuous representative exceeded the essential supremum at one point, it would exceed it on a nonempty open set. Thus the displayed L^∞ estimate holds. The proof of Theorem 2.4, equivalently [4, proposition "Local compactness"], applied to this convergent sequence of translates, also shows that W satisfies the mild formulation and is divergence free.

The identity $\Theta_t\Theta_s = \Theta_{t+s}$ follows from the definition of time translation. The map Θ_t sends the orbit to itself; the continuity proved below then implies that it sends its closure K into K. If $W_n \to W$ locally smoothly, then for every compact cylinder $B_R \times [-T, T]$ and every m,

$$\|\Theta_t W_n - \Theta_t W\|_{C^m(B_R\times[-T,T])} = \|W_n - W\|_{C^m(B_R\times[-T+t,T+t])},$$

where the right-hand side is estimated on $B_R \times [-T - |t|, T + |t|]$, one of the compact cylinders used in the locally smooth topology. The right-hand side tends to zero, so Θ_t is continuous. Applying the same argument to Θ_{-t} gives $\Theta_{-t}\Theta_t = \Theta_t\Theta_{-t} = \mathrm{Id}$ on K. Hence each Θ_t is a homeomorphism of K.

It remains to verify joint continuity of the action. Suppose $t_n \to t$ and $W_n \to W$ locally smoothly. Fix R, T, m. Choose $A > |t| + 1$ such that $|t_n| \le A$ for all large n. Then

$$\begin{aligned}\|\Theta_{t_n} W_n - \Theta_t W\|_{C^m(B_R\times[-T,T])} &\le \|\Theta_{t_n} W_n - \Theta_{t_n} W\|_{C^m(B_R\times[-T,T])} \\ &\quad + \|\Theta_{t_n} W - \Theta_t W\|_{C^m(B_R\times[-T,T])}.\end{aligned}$$

The first term is controlled by the C^m-norm of $W_n - W$ on $B_R \times [-T - A, T + A]$, hence tends to zero. The second term tends to zero because W is smooth, so each derivative $\partial_y^\alpha \partial_\tau^j W$, $|\alpha| + j \le m$, is uniformly continuous on the compact cylinder $B_R \times [-T - A, T + A]$. Thus $(t, W) \mapsto \Theta_t W$ is continuous, and the maps form a continuous group.

It remains to prove the uniform L^3 assertions. The family of translates $\{V^s : s \in \mathbb{R}\}$ satisfies (2.2) and (2.3) with the same constants because these estimates are invariant under τ-translation. Since K is the locally smooth closure of this family, the closedness under uniform L^3-tightness from [4, proposition "Closedness under uniform tightness"] applies directly. It gives the two displayed L^3 estimates for every $W \in K$, with the same L and the same tail modulus ω. No additional tightness, decay, or compact-support hypothesis is introduced. ☐

3. Compact invariant sets

Definition 3.1 (Compact invariant set)**.** A nonempty compact subset K, with respect to the locally smooth topology, of bounded mild ancient solutions is invariant if $\Theta_t K = K$ for every $t \in \mathbb{R}$. It is minimal if it has no proper nonempty compact invariant subset.

Lemma 3.2 (Continuity of the restricted translation action)**.** *Let K be a compact invariant set in the sense of Theorem 3.1. Then, for every $t \in \mathbb{R}$, $\Theta_t : K \to K$ is a homeomorphism, and the action*

$$\mathbb{R} \times K \ni (t, W) \mapsto \Theta_t W \in K$$

is jointly continuous.

Proof. The identity $\Theta_t\Theta_s = \Theta_{t+s}$ follows from the definition of time translation. Since K is invariant, $\Theta_t K = K$ for every t, so Θ_t maps K onto K and has inverse Θ_{-t}.

Fix $t \in \mathbb{R}$. If $W_n \to W$ in K locally smoothly, then for every $R, T < \infty$ and every integer $m \ge 0$,

$$\|\Theta_t W_n - \Theta_t W\|_{C^m(B_R\times[-T,T])} = \|W_n - W\|_{C^m(B_R\times[-T+t,T+t])}.$$

The interval $[-T+t, T+t]$ is contained in $[-T-|t|, T+|t|]$, so the right-hand side tends to zero by local smooth convergence. Thus Θ_t is continuous on K. Applying the same argument to Θ_{-t} proves that Θ_t is a homeomorphism.

It remains to prove joint continuity. Suppose $t_n \to t$ and $W_n \to W$ in K. Fix R, T, m, and choose $A > |t|+1$ so that $|t_n| \le A$ for all sufficiently large n. Then

$$\begin{aligned}\|\Theta_{t_n} W_n - \Theta_t W\|_{C^m(B_R\times[-T,T])} &\le \|\Theta_{t_n} W_n - \Theta_{t_n} W\|_{C^m(B_R\times[-T,T])} \\ &\quad + \|\Theta_{t_n} W - \Theta_t W\|_{C^m(B_R\times[-T,T])}.\end{aligned}$$

The first term is bounded by the C^m-norm of $W_n - W$ on $B_R \times [-T-A, T+A]$, and hence tends to zero. For the second term, each derivative $\partial_y^\alpha \partial_\tau^j W$, with $|\alpha|+j \le m$, is uniformly continuous on $\overline{B}_R \times [-T-A, T+A]$. Therefore

$$\partial_y^\alpha \partial_\tau^j W(y, \tau + t_n) \to \partial_y^\alpha \partial_\tau^j W(y, \tau + t)$$

uniformly for $(y,\tau) \in \overline{B}_R \times [-T,T]$. The second term also tends to zero, proving joint continuity. □

Proposition 3.3 (Minimal compact subsets)**.** *Every nonempty compact invariant set contains a minimal compact invariant subset.*

Proof. Let $\mathcal{F}$ be the family of nonempty compact invariant subsets of K. We order $\mathcal{F}$ by reverse inclusion:

$$A \preceq B \quad \Longleftrightarrow \quad B \subset A.$$

A maximal element for this order is exactly a minimal element for ordinary set inclusion. To apply Zorn's lemma, let $\{K_\alpha\}$ be a $\preceq$-chain. Equivalently, it is a family totally ordered by ordinary inclusion. The intersection

$$K_* := \bigcap_\alpha K_\alpha$$

is nonempty: each K_α is closed in the ambient compact set K, and finite intersections are nonempty because a finite subfamily of a totally ordered family by inclusion has a smallest member. Thus the finite intersection property and compactness of K give $K_* \neq \emptyset$. The set K_* is compact because it is closed in K.

It is invariant by Theorem 3.2. Indeed, since every member of the chain is invariant,

$$\Theta_t K_* \subset \bigcap_\alpha \Theta_t K_\alpha = K_*.$$

Applying the same inclusion with $-t$ gives

$$\Theta_{-t} K_* \subset K_*.$$

Applying Θ_t to this last inclusion and using $\Theta_t \Theta_{-t} = \mathrm{Id}$ gives the reverse inclusion

$$K_* \subset \Theta_t K_*.$$

Hence $\Theta_t K_* = K_*$. Therefore $K_* \in \mathcal{F}$. Moreover $K_* \subset K_\alpha$ for every α, so $K_\alpha \preceq K_*$ for every α; that is, K_* is an upper bound for the chain in the reverse-inclusion order. Zorn's lemma gives a maximal element of $(\mathcal{F}, \preceq)$, equivalently a nonempty compact invariant subset which is minimal under ordinary inclusion. □

Lemma 3.4 (The zero trajectory)**.** *Let K be a minimal compact invariant set. If the zero solution belongs to K, then K consists only of the zero solution.*

Proof. The zero solution satisfies $\Theta_t 0 = 0$ for every t, so the singleton $\{0\}$ is compact and invariant. If $0 \in K$, then $\{0\}$ is a nonempty compact invariant subset of K. Minimality of K therefore forces $K = \{0\}$. □

4. Invariant measures

Theorem 4.1 (Krylov–Bogolyubov averages)**.** *Let K be a compact invariant set and let $W_0 \in K$. For $T > 0$, define the probability measure*

$$\mu_T = \frac{1}{T}\int_0^T \delta_{\Theta_s W_0}\, ds$$

on K. If $T_n \to \infty$, then a subsequence of μ_{T_n} converges weakly to a Borel probability measure μ on K. Every such limit is invariant:

$$\int_K F(\Theta_t W)\, d\mu(W) = \int_K F(W)\, d\mu(W)$$

for every $t \in \mathbb{R}$ and every continuous function $F : K \to \mathbb{R}$. If K is minimal, then the support of every invariant Borel probability measure on K is equal to K.

Proof. By Theorem 3.2, the orbit map $s \mapsto \Theta_s W_0$ is continuous. The formula defining μ_T denotes the push-forward of normalized Lebesgue measure on $[0, T]$ under this orbit map,

$$[0, T] \ni s \mapsto \Theta_s W_0 \in K.$$

Thus μ_T is a Borel probability measure on K. In particular, for a Borel set $A \subset K$,

$$\mu_T(A) = \frac{1}{T}\left|\{s \in [0, T] : \Theta_s W_0 \in A\}\right|,$$

where the set on the right is Lebesgue measurable because the orbit map is continuous and A is Borel.

For $F \in C(K)$, the map

$$s \mapsto F(\Theta_s W_0)$$

is continuous, hence Borel and bounded on every finite interval. Therefore μ_T is the Borel probability measure characterized by

$$\int_K F(W)\, d\mu_T(W) = \frac{1}{T}\int_0^T F(\Theta_s W_0)\, ds.$$

Since K is compact and metrizable, $C(K)$ is separable. By the Riesz representation theorem, Borel probability measures on K are the positive norm-one elements of $C(K)^*$. Banach–Alaoglu gives weak-* compactness of the unit ball of $C(K)^*$, and separability of $C(K)$ makes this compact set metrizable on bounded subsets. Consequently every sequence of Borel probability measures on K has a weakly convergent subsequence [1]. Applying this to μ_{T_n}, choose a subsequence which converges weak-* in $C(K)^*$ to a bounded linear functional L on $C(K)$. In concrete terms,

$$\int_K F\, d\mu_{T_{n_k}} \to L(F) \qquad \text{for every } F \in C(K).$$

For every $F \in C(K)$ with $F \geq 0$, one has

$$L(F) = \lim_{k\to\infty} \int_K F\, d\mu_{T_{n_k}} \geq 0,$$

and, for the constant function 1,

$$L(1) = \lim_{k\to\infty} \mu_{T_{n_k}}(K) = 1.$$

The Riesz representation theorem therefore identifies L with integration against a Borel probability measure μ on K. Thus the subsequence converges weakly to μ.

Let F be continuous and fix $t > 0$. Then

$$\int_K F(\Theta_t W)\, d\mu_T(W) - \int_K F(W)\, d\mu_T(W) = \frac{1}{T}\int_T^{T+t} F(\Theta_s W_0)\, ds - \frac{1}{T}\int_0^t F(\Theta_s W_0)\, ds.$$

The absolute value of the right-hand side is at most

$$\frac{2t}{T}\|F\|_{L^\infty(K)}.$$

The displayed equality follows by the change of variables $r = s + t$ in the first time average. Since $W \mapsto F(\Theta_t W)$ is continuous on K, weak convergence of $\mu_{T_{n_k}}$ to μ gives convergence of both integrals on the left along the chosen subsequence. Letting $T = T_{n_k} \to \infty$ gives invariance for $t > 0$. If $t < 0$, apply the positive-time result with $-t$ to the continuous function $F \circ \Theta_t$. Since $\Theta_{-t}\Theta_t$ is the identity, this gives the asserted identity for negative t as well. For $t = 0$ the identity is trivial. Equivalently, $(\Theta_t)_{\#}\mu = \mu$ as Borel measures for every $t \in \mathbb{R}$. The last equivalence follows from the uniqueness part of the Riesz representation theorem, because the two probability measures have the same integral against every $F \in C(K)$.

If K is minimal and μ is invariant, let $\operatorname{supp}\mu$ denote the closed support, namely the set of all points $W \in K$ such that every open neighborhood of W has positive μ-measure. Its complement is open: if $W \notin \operatorname{supp}\mu$, then some open neighborhood U of W has $\mu(U) = 0$, and the same U is a zero-measure neighborhood for each of its points. Hence $\operatorname{supp}\mu$ is closed. This set is nonempty: if it were empty, then every $W \in K$ would have an open neighborhood U_W with $\mu(U_W) = 0$; compactness would give a finite subcover of K, and hence $\mu(K) = 0$, contradicting $\mu(K) = 1$. The support is compact because it is closed in the compact space K.

It is also invariant. Indeed, if $W \in \operatorname{supp}\mu$ and U is an open neighborhood of $\Theta_t W$, then $\Theta_{-t}U$ is an open neighborhood of W, hence $\mu(\Theta_{-t}U) > 0$. Since $(\Theta_t)_{\#}\mu = \mu$,

$$\mu(U) = \mu(\Theta_{-t}U) > 0.$$

It follows that $\Theta_t W \in \operatorname{supp}\mu$, so $\Theta_t(\operatorname{supp}\mu) \subset \operatorname{supp}\mu$. Applying the same argument to $-t$ gives the opposite inclusion. Hence the support is invariant. The support is therefore a nonempty compact invariant subset of K, and minimality forces it to be K. □

Corollary 4.2 (Nonzero support). *Let K be a minimal compact invariant set containing at least one nonzero trajectory. Then the zero trajectory is not in K, and every invariant Borel probability measure on K has support equal to K. In particular, no such measure is supported at the zero solution.*

Proof. The first assertion follows from Theorem 3.4. The support statement is the last assertion of Theorem 4.1. □

5. The stationary statistical identity

The next theorem records the exact weak identity satisfied by invariant measures. The test fields are solenoidal, so the pressure is eliminated in the usual distributional formulation.

Theorem 5.1 (Stationary statistical form). *Let K be a compact invariant set of bounded mild ancient solutions of (1.1), and let μ be an invariant Borel probability measure on K. Then, for* *every solenoidal test field* $\phi \in C_c^\infty(\mathbb{R}^3;\mathbb{R}^3)$,

$$
\begin{aligned}
0 = \int_K \Bigg[\int_{\mathbb{R}^3} W(y,0) \cdot \left(\Delta\phi(y) + \phi(y) + \frac{1}{2} y \cdot \nabla\phi(y) \right) dy \\
+ \int_{\mathbb{R}^3} W_i(y,0) W_j(y,0) \partial_j \phi_i(y)\, dy \Bigg] d\mu(W).
\end{aligned} \tag{5.1}
$$

Proof. For such a test field define

$$
F(W) = \int_{\mathbb{R}^3} W(y,0) \cdot \phi(y)\, dy.
$$

If $W_n \to W$ locally smoothly, then $W_n(\cdot,0) \to W(\cdot,0)$ uniformly on the compact support of ϕ. Hence $F(W_n) \to F(W)$, so $F \in C(K)$. Invariance gives

$$
\int_K F(\Theta_h W)\, d\mu(W) = \int_K F(W)\, d\mu(W) \qquad h \in \mathbb{R}.
$$

Therefore, for $0 < |h| \le 1$,

$$
\int_K \frac{F(\Theta_h W) - F(W)}{h}\, d\mu(W) = 0.
$$

Choose R such that $\operatorname{supp}\phi \subset B_R$. Because

$$
W \mapsto \|\partial_\tau W\|_{L^\infty(\overline{B}_R \times [-1,1])}
$$

is continuous on K and K is compact, the quantity

$$
M_\phi := \sup_{W \in K} \|\partial_\tau W\|_{L^\infty(\overline{B}_R \times [-1,1])}
$$

is finite. For $0 < |h| \le 1$, the fundamental theorem of calculus in the τ-variable gives, pointwise for $y \in \operatorname{supp}\phi$,

$$
\frac{F(\Theta_h W) - F(W)}{h} = \int_{\mathbb{R}^3} \left(\frac{1}{h} \int_0^h \partial_\tau W(y,s)\, ds \right) \cdot \phi(y)\, dy.
$$

The absolute value is bounded by

$$
|\operatorname{supp}\phi|\, \|\phi\|_{L^\infty} M_\phi,
$$

which is independent of W and h. For each fixed W, smoothness in τ gives convergence of the inner average to $\partial_\tau W(y,0)$, uniformly for $y \in \operatorname{supp}\phi$. The integrands as functions of W are Borel, because they are continuous with respect to the locally smooth topology. Dominated convergence on (K,μ) gives

$$
\int_K \int_{\mathbb{R}^3} \partial_\tau W(y,0) \cdot \phi(y)\, dy\, d\mu(W) = 0.
$$

For each $W \in K$, Theorem 2.3 gives a smooth local pressure on a cylinder containing $\operatorname{supp}\phi \times \{0\}$. We write it as P on that cylinder. Pairing (1.1) at time 0 with ϕ, in the sense of distributions and then classically on the compact support of ϕ, gives

$$
\begin{aligned}
\int_{\mathbb{R}^3} \partial_\tau W \cdot \phi\, dy = \int_{\mathbb{R}^3} \Delta W \cdot \phi\, dy - \frac{1}{2} \int_{\mathbb{R}^3} W \cdot \phi\, dy - \frac{1}{2} \int_{\mathbb{R}^3} (y \cdot \nabla W) \cdot \phi\, dy \\
- \int_{\mathbb{R}^3} (W \cdot \nabla) W \cdot \phi\, dy - \int_{\mathbb{R}^3} \nabla P \cdot \phi\, dy.
\end{aligned}
$$

All terms in this display are evaluated at $\tau = 0$, and all integrations by parts below are over compact subsets of the pressure cylinder, because ϕ is compactly supported. The pressure term vanishes:

$$\int_{\mathbb{R}^3} \nabla P \cdot \phi \, dy = -\int_{\mathbb{R}^3} P \operatorname{div} \phi \, dy = 0.$$

Two integrations by parts give

$$\int_{\mathbb{R}^3} \Delta W \cdot \phi \, dy = \int_{\mathbb{R}^3} W \cdot \Delta \phi \, dy.$$

Using $\operatorname{div} W = 0$, the convection term is

$$-\int_{\mathbb{R}^3} (W \cdot \nabla) W \cdot \phi \, dy = -\int_{\mathbb{R}^3} W_j \partial_j W_i \, \phi_i \, dy = \int_{\mathbb{R}^3} W_i W_j \partial_j \phi_i \, dy.$$

Here and below repeated spatial indices are summed over $\{1, 2, 3\}$. For the drift term,

$$\int_{\mathbb{R}^3} (y \cdot \nabla W) \cdot \phi \, dy = \int_{\mathbb{R}^3} y_j \partial_j W_i \, \phi_i \, dy = -\int_{\mathbb{R}^3} W_i \partial_j (y_j \phi_i) \, dy.$$

Since $\partial_j y_j = 3$, this equals

$$-3 \int_{\mathbb{R}^3} W \cdot \phi \, dy - \int_{\mathbb{R}^3} W \cdot (y \cdot \nabla \phi) \, dy.$$

Therefore

$$-\frac{1}{2} \int_{\mathbb{R}^3} W \cdot \phi \, dy - \frac{1}{2} \int_{\mathbb{R}^3} (y \cdot \nabla W) \cdot \phi \, dy = \int_{\mathbb{R}^3} W \cdot \phi \, dy + \frac{1}{2} \int_{\mathbb{R}^3} W \cdot (y \cdot \nabla \phi) \, dy.$$

Combining these identities, and evaluating W at $\tau = 0$, gives for each W

$$\begin{aligned} \int_{\mathbb{R}^3} \partial_\tau W(y, 0) \cdot \phi(y) \, dy = \int_{\mathbb{R}^3} W(y, 0) \cdot \left(\Delta \phi(y) + \phi(y) + \frac{1}{2} y \cdot \nabla \phi(y) \right) dy \\ + \int_{\mathbb{R}^3} W_i(y, 0) W_j(y, 0) \partial_j \phi_i(y) \, dy. \end{aligned}$$

The right-hand side is a continuous function of $W \in K$, by local smooth convergence on $\operatorname{supp} \phi \times \{0\}$. It is bounded on K because the seminorm

$$W \mapsto \|W(\cdot, 0)\|_{L^\infty(\operatorname{supp} \phi)}$$

is bounded on the compact set K, and the displayed expression is bounded by a constant depending only on ϕ, $|\operatorname{supp} \phi|$, and this uniform local L^∞ bound. Therefore it is μ-integrable. Substituting the pointwise identity for $\int \partial_\tau W(y, 0) \cdot \phi(y) \, dy$ into the averaged identity

$$\int_K \int_{\mathbb{R}^3} \partial_\tau W(y, 0) \cdot \phi(y) \, dy \, d\mu(W) = 0$$

gives (5.1). □

Theorem 5.2 (Barycenter and covariance)**.** *Let K and μ be as in Theorem 5.1. Define the barycenter*

$$\overline{W}(y) = \int_K W(y, 0) \, d\mu(W)$$

and the quadratic covariance

$$C_{ij}(y) = \int_K W_i(y, 0) W_j(y, 0) \, d\mu(W) - \overline{W}_i(y) \overline{W}_j(y).$$

Then $\overline{W}$ and C are C^∞_{loc}, and $\overline{W}$ is solenoidal. Moreover, for every solenoidal $\phi \in C_c^\infty(\mathbb{R}^3;\mathbb{R}^3)$,

$$\int_{\mathbb{R}^3} \overline{W}\cdot\left(\Delta\phi+\phi+\frac{1}{2}y\cdot\nabla\phi\right)dy+\int_{\mathbb{R}^3}\overline{W}_i\overline{W}_j\partial_j\phi_i\,dy+\int_{\mathbb{R}^3}C_{ij}\partial_j\phi_i\,dy=0. \tag{5.2}$$

Consequently, $\overline{W}$ is a weak stationary solution of (1.1), in the pressure-free formulation against solenoidal test fields, if and only if

$$\int_{\mathbb{R}^3} C_{ij}(y)\partial_j\phi_i(y)\,dy=0 \tag{5.3}$$

for every solenoidal $\phi\in C_c^\infty(\mathbb{R}^3;\mathbb{R}^3)$.

Proof. Fix a compact ball B_R and a multiindex α. Compactness of K in the locally smooth topology gives

$$M_{R,\alpha}:=\sup_{W\in K}\sup_{y\in B_R}|\partial_y^\alpha W(y,0)|<\infty.$$

For each y, the evaluation map $W\mapsto W(y,0)$ is continuous from K into the finite-dimensional space $\mathbb{R}^3$. Its image is compact, hence bounded and Borel measurable. The barycenter is therefore well defined pointwise as a finite vector-valued Bochner integral; equivalently, since the range is $\mathbb{R}^3$, it is obtained by integrating the three scalar components separately. We next justify the spatial differentiation under the integral sign.

Let e_ℓ be a coordinate vector and let $y\in B_R$. If $|h|$ is small enough that the segment $\{y+\theta h e_\ell : 0\le\theta\le 1\}$ is contained in a fixed slightly larger ball $B_{R'}$, then

$$\frac{\partial_y^\alpha W(y+he_\ell,0)-\partial_y^\alpha W(y,0)}{h}=\int_0^1\partial_\ell\partial_y^\alpha W(y+\theta he_\ell,0)\,d\theta.$$

The right-hand side is bounded uniformly in $W\in K$ by $M_{R',\alpha+e_\ell}$, and it converges pointwise in W, as $h\to 0$, to $\partial_\ell\partial_y^\alpha W(y,0)$. Dominated convergence on (K,μ) therefore gives

$$\partial_\ell\int_K\partial_y^\alpha W(y,0)\,d\mu(W)=\int_K\partial_\ell\partial_y^\alpha W(y,0)\,d\mu(W).$$

Starting from $\alpha=0$ and iterating this coordinate argument over the components of a multiindex yields, for $y\in B_R$,

$$\partial_y^\alpha\overline{W}(y)=\int_K\partial_y^\alpha W(y,0)\,d\mu(W),$$

with the case $\alpha=0$ understood as the definition of $\overline{W}$. The last display also proves continuity of $\partial_y^\alpha\overline{W}$: if $y_n\to y$ in B_R, then $\partial_y^\alpha W(y_n,0)\to\partial_y^\alpha W(y,0)$ pointwise in W, while the integrands are dominated by $M_{R,\alpha}$. Hence $\overline{W}\in C^\infty_{\mathrm{loc}}$.

For the second moment, first observe that for each fixed y the map

$$W\mapsto W_i(y,0)W_j(y,0)$$

is continuous on K, hence Borel measurable. It is bounded because the scalar evaluations $W\mapsto W_i(y,0)$ and $W\mapsto W_j(y,0)$ have compact images. Hence the scalar Lebesgue integral

$$M_{ij}(y):=\int_K W_i(y,0)W_j(y,0)\,d\mu(W)$$

is well defined. The same differentiation-under-the-integral argument applies to M_{ij}. Indeed, on B_R, Leibniz' rule writes $\partial_y^\alpha(W_iW_j)$ as a finite sum of products of derivatives of W_i and W_j of order at most $|\alpha|$. Each factor is uniformly bounded on $K\times B_R$ by compactness of K, and hence every product has an integrable bound independent of W. Repeating the coordinate difference-quotient argument used above gives

$$\partial_y^\alpha M_{ij}(y) = \int_K \partial_y^\alpha \big(W_i(y,0)W_j(y,0)\big)\, d\mu(W).$$

Since $C_{ij} = M_{ij} - \overline{W}_i\overline{W}_j$, the covariance tensor also belongs to C^∞_{loc}.

To prove that $\overline{W}$ is solenoidal, take $\eta \in C_c^\infty(\mathbb{R}^3)$. Since every $W \in K$ is divergence free,

$$\int_{\mathbb{R}^3} W(y,0)\cdot\nabla\eta(y)\, dy = 0.$$

The map

$$(W,y) \mapsto W(y,0)\cdot\nabla\eta(y)$$

is continuous on $K\times\operatorname{supp}\eta$ and vanishes outside the compact support of η in the y-variable. It is therefore Borel and absolutely bounded by

$$\sup_{W\in K}\|W(\cdot,0)\|_{L^\infty(\operatorname{supp}\eta)}\|\nabla\eta\|_{L^\infty}\mathbf{1}_{\operatorname{supp}\eta}(y),$$

which is integrable with respect to $\mu\otimes dy$. Fubini's theorem gives

$$\int_{\mathbb{R}^3}\overline{W}(y)\cdot\nabla\eta(y)\, dy = \int_K\int_{\mathbb{R}^3} W(y,0)\cdot\nabla\eta(y)\, dy\, d\mu(W) = 0.$$

Hence $\operatorname{div}\overline{W} = 0$ in distributions, and therefore classically because $\overline{W}$ is smooth.

Now apply Theorem 5.1. The linear term may be passed through the K-integral by Fubini–Tonelli, since its absolute value is bounded by

$$|\operatorname{supp}\phi|\sup_{W\in K}\|W(\cdot,0)\|_{L^\infty(\operatorname{supp}\phi)}\left\|\Delta\phi+\phi+\frac{1}{2}y\cdot\nabla\phi\right\|_{L^\infty},$$

which is finite:

$$\int_K\int_{\mathbb{R}^3} W\cdot\left(\Delta\phi+\phi+\frac12 y\cdot\nabla\phi\right)dy\, d\mu(W) = \int_{\mathbb{R}^3}\overline{W}\cdot\left(\Delta\phi+\phi+\frac12 y\cdot\nabla\phi\right)dy.$$

For the nonlinear term, local boundedness on $\operatorname{supp}\phi$ gives the finite bound

$$|\operatorname{supp}\phi|\sup_{W\in K}\|W(\cdot,0)\|^2_{L^\infty(\operatorname{supp}\phi)}\|\nabla\phi\|_{L^\infty},$$

so Fubini–Tonelli applies again. By the definition of C_{ij},

$$\int_K W_iW_j\, d\mu = \overline{W}_i\overline{W}_j + C_{ij}.$$

Therefore

$$\int_K\int_{\mathbb{R}^3} W_iW_j\partial_j\phi_i\, dy\, d\mu(W) = \int_{\mathbb{R}^3}\overline{W}_i\overline{W}_j\partial_j\phi_i\, dy + \int_{\mathbb{R}^3} C_{ij}\partial_j\phi_i\, dy.$$

Substitution in (5.1) gives (5.2).

Finally, we spell out the pressure-free weak stationary formulation used here. Since $\overline{W}$ is smooth and solenoidal, testing the stationary self-similar equation

$$(\overline{W}\cdot\nabla)\overline{W} + \nabla\overline{P} = \Delta\overline{W} - \frac12(\overline{W} + y\cdot\nabla\overline{W}), \qquad \operatorname{div}\overline{W} = 0,$$

against a solenoidal $\phi\in C_c^\infty(\mathbb{R}^3;\mathbb{R}^3)$ eliminates the pressure term:

$$\int_{\mathbb{R}^3}\nabla\overline{P}\cdot\phi\, dy = -\int_{\mathbb{R}^3}\overline{P}\operatorname{div}\phi\, dy = 0.$$

The same compactly supported integrations by parts used in Theorem 5.1 then give the pressure-free identity

$$\int_{\mathbb{R}^3} \overline{W} \cdot \left(\Delta\phi + \phi + \frac{1}{2} y \cdot \nabla\phi \right) dy + \int_{\mathbb{R}^3} \overline{W}_i \overline{W}_j \partial_j \phi_i \, dy = 0$$

for every solenoidal $\phi \in C_c^\infty$. Conversely, this is exactly what is meant in this theorem by being stationary in the pressure-free weak formulation: no pressure is asserted or needed in the equivalence. Comparing this identity with (5.2) shows that the barycenter is stationary in that weak sense if and only if the covariance term satisfies (5.3) for every such test field. ☐

Corollary 5.3 (Stationary support gives a stationary profile)**.** *Let K be a compact invariant set of bounded mild ancient solutions. Assume that there is $L < \infty$ such that every $W \in K$ satisfies*

$$\sup_{\tau\in\mathbb{R}} \|W(\tau)\|_{L^3(\mathbb{R}^3)} \le L.$$

Let μ be an invariant probability measure on K. If μ gives positive mass to the set of stationary trajectories, then K contains a stationary trajectory $W(y,\tau) = w(y)$ whose profile satisfies $w \in L^3(\mathbb{R}^3)$. If, in addition, the stationary trajectory is nonzero, this contradicts the stationary self-similar theorem of Nečas–Růžička–Šverák.

Proof. Let

$$\mathcal{S} = \{W \in K : \Theta_t W = W \text{ for every } t \in \mathbb{R}\}.$$

For $W \in K$, it suffices to check equality for rational t. Indeed, if $\Theta_q W = W$ for every $q \in \mathbb{Q}$, then for any $t \in \mathbb{R}$ choose $q_n \in \mathbb{Q}$ with $q_n \to t$. Joint continuity of the action, proved in Theorem 3.2, gives $\Theta_{q_n} W \to \Theta_t W$ in K, while $\Theta_{q_n} W = W$ for all n. Since the locally smooth topology is Hausdorff, $\Theta_t W = W$. Hence

$$\mathcal{S} = \bigcap_{t\in\mathbb{Q}} \{W \in K : \Theta_t W = W\}.$$

Each set in the intersection is closed. Indeed, for fixed t the map

$$K \ni W \mapsto (\Theta_t W, W) \in K \times K$$

is continuous, and $\{W \in K : \Theta_t W = W\}$ is the inverse image of the diagonal in $K \times K$. The diagonal is closed because the locally smooth topology is Hausdorff. Thus $\mathcal{S}$ is a countable intersection of closed sets and is Borel. If $\mu(\mathcal{S}) > 0$, then $\mathcal{S}$ is nonempty: otherwise it would be the empty Borel set and would have measure zero. Choose $W \in \mathcal{S}$.

For this W, the identity $\Theta_t W = W$ for every t means

$$W(y, \tau + t) = W(y, \tau)$$

for all y, τ, t. Taking $\tau = 0$ gives $W(y,t) = W(y,0)$, so W is independent of τ. Write $w(y) = W(y,0)$. By Theorem 2.3, on every compact ball there is a smooth pressure representative for W, unique up to addition of a function of time. Since $\partial_\tau W = 0$, evaluating (1.1) at $\tau = 0$ gives, on every compact ball, the stationary self-similar equation

$$(w\cdot\nabla)w + \nabla p = \Delta w - \frac{1}{2}(w + y\cdot\nabla w), \qquad \operatorname{div} w = 0.$$

On overlaps of two balls the two pressure gradients agree, because both equal the same smooth vector field

$$\Delta w - \frac{1}{2}(w + y\cdot\nabla w) - (w\cdot\nabla)w.$$

We now patch the local pressures. Let $B_m = B_m(0)$, $m \geq 1$, and choose local pressures $p_m \in C^\infty(B_m)$ at time $\tau = 0$. On $B_m \cap B_{m+1} = B_m$ one has

$$\nabla(p_{m+1} - p_m) = 0.$$

Since B_m is connected, $p_{m+1} - p_m$ is constant on B_m. Replacing p_{m+1} by p_{m+1} minus that constant, and proceeding inductively, we may assume $p_{m+1} = p_m$ on B_m for every m. The compatible family then defines a smooth global pressure $p \in C^\infty_{\mathrm{loc}}(\mathbb{R}^3)$ by $p = p_m$ on B_m. Hence

$$(w \cdot \nabla)w + \nabla p = \Delta w - \frac{1}{2}(w + y \cdot \nabla w), \qquad \operatorname{div} w = 0$$

holds classically on every compact subset of $\mathbb{R}^3$, and therefore globally in $\mathcal{D}'(\mathbb{R}^3)$. The assumed uniform bound gives $\|w\|_{L^3(\mathbb{R}^3)} = \|W(\cdot, 0)\|_{L^3(\mathbb{R}^3)} \leq L$. The theorem of Nečas–Růžička–Šverák [6] applies to smooth stationary self-similar solutions in $L^3(\mathbb{R}^3)$ and forces $w = 0$. Therefore any stationary trajectory in K satisfying the displayed L^3 bound is the zero trajectory, since $W(y, \tau) = w(y)$ for all τ. In particular a nonzero stationary trajectory would contradict that theorem. This is the same stationary-rigidity interface recorded in [4, corollary "Interface with stationary rigidity"]; the preceding paragraphs only identify the stationary element inside the present invariant hull. □

6. Conclusion

Bounded ancient solutions generate compact trajectory hulls in the locally smooth topology, and uniform L^3-tightness is stable on those hulls. Compact invariant subsets support invariant probability measures obtained by time averaging. These measures satisfy an exact stationary statistical weak form of the self-similar Navier–Stokes equations. The obstruction to turning the statistical identity into a single stationary solution is precisely the quadratic covariance in Theorem 5.2. A later rigidity argument must show, under additional hypotheses, that compact nonzero ancient dynamics place mass on stationary trajectories or otherwise make this covariance vanish.

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